

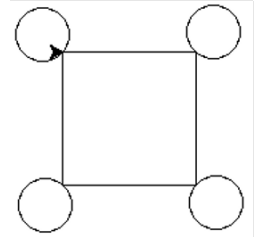
Information Processing 1 — Week 5 exercises

Complete each exercise on your own laptop before the end of the exercise class. When you finish an exercise, immediately ask an instructor to check your answer. Answers checked after the end of the class will have a 50% penalty.

If you are stuck or have any questions then raise your hand and an instructor will try to help you.

1 Drawing geometric shapes [2 points]

During the class we created a function called `circle(radius)`. Use that to write another function called `squareCircles(length, radius)` that draws a square with sides of the given `length` and four circles of the given `radius` that just touch the corners of the square as shown in the figure.



Hint: Develop your function *incrementally*. While drawing a square, draw a circle at the end of each edge. Then figure out how to orient the circles correctly. Alternatively, start by drawing just the first circle and the first line. When they look correct, repeat the code four times to draw the entire figure.

▷▷ Ask an instructor to check and record your work **now** (before you run out of time).

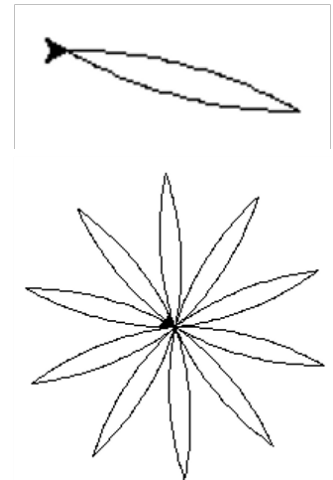
2 Drawing flowers [4 points]

During the class we created a function called `arc(radius, angle)`.

Part 1 Use the `arc(radius, angle)` function to write another function called `petal(radius, angle)` that draws a single flower 'petal'.

Hint: Ensure that the final position and final heading of the pen, after drawing the petal, are the same as the starting position and heading. In other words, if you call `petal` twice it should overwrite itself.

Part 2 Write a function called `flower(radius, angle, npetals)` that draws a flower with `npetals` petals each of which has the given `radius` and `angle`.

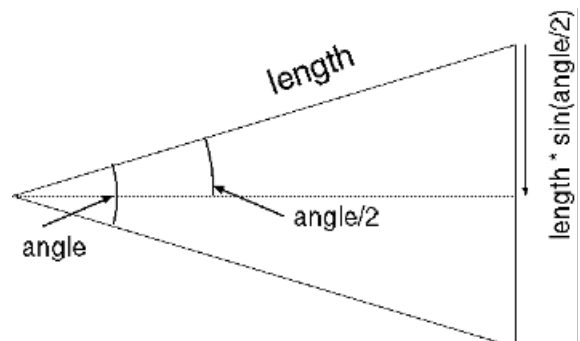


▷▷ Ask an instructor to check and record your work **now** (before you run out of time).

3 Drawing pizzas [4 points]

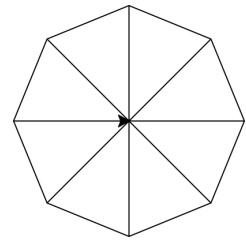
Part 1 Write a function `triangle(length, angle)` that draws an *isosceles* triangle whose equal sides are the given `length` and separated by the given `angle`.

Hints: The problem is easier to solve if you think of the triangle as two right-angled triangles back-to-back, both having a hypotenuse of the given `length`. You can then use `length * math.sin(angle / 2)` to calculate half the length of the base of the triangle. Don't forget that `angle` must be converted to radians (not shown here) before you use it in the `sin()` function.



Part 2 Write a function `pizza(length, number)` that draws `number` copies of an isosceles triangle, whose equal edges are the given `length`, whose apexes all touch in the centre of the drawing, and whose bases form a 360 degree polygon.

Hints: Use `triangle()` to draw each segment. Calculate the `angle` based on the `number` of segments. Rotated each triangle from the previous by `angle` degrees to form the pattern shown here.



▷ Ask an instructor to check and record your work **now** (before you run out of time).

4 Challenges [1 bonus point]

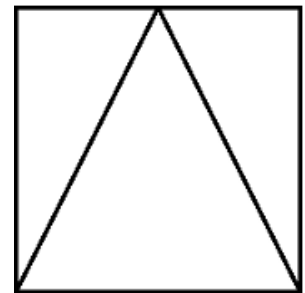
Earn a bonus point by completing these challenges and providing a thoughtful answer to the final question.

4.1 Precisely explaining a pattern — to Python

In Week 1 you practised describing how to draw this pattern, very precisely, to your partner using a common native language.

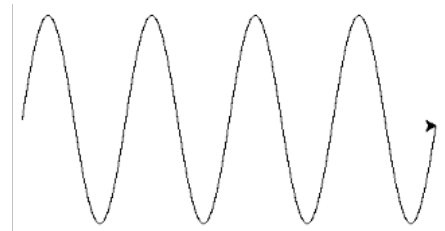
This week describe precisely how to draw this pattern to your computer using the Python language. Create a function to perform the drawing. Parameterise the function with the length of the sides of the square.

Hint: One way to do this is to calculate the length of the triangle's sides and then re-use your `triangle()` function from earlier.



4.2 Moving the 'pen' with goto

Write a function that uses `goto(x, y)` to plot two cycles of a sine wave (angles in the range -2π to 2π) along the X axis (from $x = -200$ to $x = 200$). Multiply the value of the sine function by 100 to scale it nicely. Hint: use `up()` and `down()` to turn drawing off and on.



4.3 Plotting an arbitrary function

Instead of plotting the sine function, modify your program to plot an arbitrary function `f(angle)`. Test your program first with `f(angle)` equal to `sin(angle)`, which should produce the same result as the previous question.

Set $f(a) = \sin(a) + \frac{1}{3} \sin(3a)$ and plot the result.

Set $f(a) = \sin(a) + \frac{1}{3} \sin(3a) + \frac{1}{5} \sin(5a)$ and plot the result.

Use a `for` loop to extend this series so that

$$f(a) = \sin(a) + \frac{1}{3} \sin(3a) + \frac{1}{5} \sin(5a) + \dots + \frac{1}{97} \sin(97a) + \frac{1}{99} \sin(99a)$$

and plot the result.

What does this tell us about square waves from the point of view of communications engineering, signal measurement, or digital signal processing (e.g., digital-to-analogue conversion)?

▷▷ Ask an instructor to check your work and award you a bonus point.